

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12413694

Kind Code

B2

Date of Patent

September 09, 2025

Inventor(s)

Tu; Tsung-Wei et al.

Method and device for adjusting depth of stereoscopic image

Abstract

A method for adjusting a depth of a stereoscopic image includes the following. A first disparity correction amount is obtained according to a distance between centers of two lenses of a stereoscopic camera and an interpupillary distance (IPD) of a first observer. A second disparity correction amount is obtained according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity. A third disparity correction amount is obtained according to the IPD of the first observer and an IPD of a second observer. A stereoscopic image of an object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is transmitted to the stereoscopic display. A device for adjusting a depth of a stereoscopic image is also provided.

Inventors: Tu; Tsung-Wei (New Taipei, TW), Huang; Chao-Shih (New Taipei, TW)

Applicant: Acer Incorporated (New Taipei, TW)

Family ID: 88586061

Assignee: Acer Incorporated (New Taipei, TW)

Appl. No.: 18/363716

Filed: August 01, 2023

Prior Publication Data

Document Identifier

Publication Date

US 20240223740 A1

Jul. 04, 2024

Foreign Application Priority Data

TW

111150471

Dec. 28, 2022

Publication Classification

Int. Cl.: H04N13/128 (20180101); H04N13/368 (20180101); H04N13/371 (20180101);
H04N13/398 (20180101)

U.S. Cl.:

CPC H04N13/128 (20180501); H04N13/368 (20180501); H04N13/371 (20180501);
H04N13/398 (20180501);

Field of Classification Search

CPC: H04N (13/128); H04N (13/368); H04N (13/371); H04N (13/398)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5065236	12/1990	Diner	348/42	G02B 30/22
5825456	12/1997	Tabata	348/51	H04N 13/383
5936663	12/1998	Tabata	348/E13.067	H04N 13/111
6798406	12/2003	Jones	348/42	H04N 13/296
9445072	12/2015	Stefanoski	N/A	H04N 13/122
9549167	12/2016	Ushiki	N/A	H04N 13/178
9612657	12/2016	Bertram	N/A	H04N 13/156
9641826	12/2016	Carlson	N/A	H04N 13/275
10595001	12/2019	Kim	N/A	H04N 13/122
10616567	12/2019	Clemens	N/A	G06T 15/20
10656596	12/2019	Callagy	N/A	G03H 1/0005
10855965	12/2019	Jiao	N/A	H04N 13/383
10922870	12/2020	Vaganov	N/A	G06T 15/02
2004/0233275	12/2003	Tomita	348/42	G02B 30/25
2004/0252374	12/2003	Saishu	359/462	H04N 13/305
2005/0001833	12/2004	Kim	348/E13.058	H04N 13/275
2005/0089212	12/2004	Mashitani	348/E13.059	H04N 13/366
2005/0190258	12/2004	Siegel	348/42	H04N 13/111
2005/0271303	12/2004	Simpson	348/E13.067	G02B 30/34
2007/0188667	12/2006	Schwerdtner	348/E13.05	H04N 13/305
2008/0049100	12/2007	Lipton	348/43	H04N 13/144
2008/0246757	12/2007	Ito	348/E13.008	G06T 15/10
2009/0219283	12/2008	Hendrickson	345/473	H04N 13/128
2009/0295790	12/2008	Pockett	345/419	H04N 13/239
2010/0039502	12/2009	Robinson	348/47	H04N 13/246
2010/0142801	12/2009	Koppal	382/154	H04N 13/10
2010/0142924	12/2009	Yamashita	348/42	G11B 27/10
2010/0220178	12/2009	Takahashi	348/54	H04N 13/10
2010/0253680	12/2009	Kobayashi	345/419	H04N 13/189
2010/0272417	12/2009	Nagasawa	386/239	H04N 13/183
2010/0309296	12/2009	Harrold	348/54	H04N 13/317
2010/0328533	12/2009	Mitani	348/521	H04N 13/356
2011/0012995	12/2010	Watanabe	348/51	H04N 13/239
2011/0050864	12/2010	Bond	348/42	H04N 13/261

2011/0063421	12/2010	Kubota	348/E13.001	H04N 13/10
2011/0090215	12/2010	Ohta	345/419	H04N 13/275
2011/0157313	12/2010	Chiang	348/51	H04N 21/4782
2011/0161843	12/2010	Bennett	715/760	H04N 21/235
2011/0175907	12/2010	Tokuda	345/419	H04N 13/128
2011/0181593	12/2010	Hirai	382/190	H04N 13/128
2011/0211815	12/2010	Yamashita	386/E5.028	H04N 13/128
2011/0229012	12/2010	Singhal	382/154	H04N 13/111
2011/0242286	12/2010	Pace	348/47	H04N 13/133
2011/0242296	12/2010	Ishihara	348/54	H04N 13/183
2011/0292183	12/2010	Tajiri	348/E13.074	H04N 13/189
2011/0305443	12/2010	Sasaki	386/E5.028	H04N 13/183
2012/0002014	12/2011	Walsh	348/47	G06T 19/006
2012/0038745	12/2011	Yu	348/46	H04N 13/183
2012/0062556	12/2011	Yamamoto	345/419	H04N 13/366
2012/0086711	12/2011	Lee	345/419	H04N 13/183
2012/0086714	12/2011	Yeon	345/419	H04N 13/361
2012/0139906	12/2011	Zhang	345/419	G06T 19/006
2012/0146894	12/2011	Yang	345/156	H04N 13/279
2012/0169730	12/2011	Inoue	345/419	H04N 13/128
2012/0182406	12/2011	Woo	359/464	H04N 13/398
2012/0229628	12/2011	Ishiyama	348/135	H04N 13/239
2012/0236002	12/2011	Bi	345/427	G06T 15/30
2012/0249532	12/2011	Kawada	345/419	H04N 13/144
2012/0256912	12/2011	Yamaguchi	345/419	H04N 13/128
2012/0274629	12/2011	Baek	345/419	H04N 13/128
2012/0306860	12/2011	Hatta	345/419	H04N 13/366
2012/0307023	12/2011	Freiburg	348/51	H04N 13/128
2012/0320047	12/2011	Yanagita	345/419	G06F 3/0418
2013/0010059	12/2012	Kobayashi	348/E13.064	G06T 19/20
2013/0021458	12/2012	Inoue	348/56	G02B 30/24
2013/0038606	12/2012	Ushiki	348/46	H04N 13/351
2013/0050196	12/2012	Takashima	345/419	H04N 13/376
2013/0050197	12/2012	Oosawa	345/419	H04N 13/327
2013/0063576	12/2012	Okubo	348/51	G02B 30/33
2013/0070052	12/2012	Yamashita	348/43	H04N 13/178
2013/0107014	12/2012	Masuda	348/47	H04N 13/239
2013/0148944	12/2012	Ando	386/278	G11B 27/034
2013/0162641	12/2012	Zhang	345/419	H04N 13/128
2013/0165186	12/2012	Choi	455/566	H04N 13/128
2013/0169753	12/2012	Lee	348/43	H04N 13/178
2013/0170737	12/2012	Arita	382/154	H04N 13/239
2013/0187852	12/2012	Ebina	345/158	H04N 13/398
2013/0194253	12/2012	Ohmi	345/419	H04N 13/128
2013/0215106	12/2012	Yamaguchi	345/419	H04N 13/128
2013/0215237	12/2012	Inoue	345/419	H04N 13/204
2013/0229407	12/2012	Chiu	345/419	H04N 13/31
2013/0242068	12/2012	Horiuchi	348/55	G02B 27/0093
2013/0268882	12/2012	Roh	715/782	G06F 3/04815
2013/0293446	12/2012	Yoo	345/6	H04N 13/356
2014/0078273	12/2013	Ono	348/54	G03B 35/26

2014/0085442	12/2013	Ono	348/54	H04N 13/337
2014/0143733	12/2013	Jung	715/848	H04N 13/261
2014/0146143	12/2013	Park	348/51	H04N 13/351
2014/0300711	12/2013	Kroon	348/51	H04N 13/307
2014/0354796	12/2013	Hein	359/385	G02B 21/086
2015/0036211	12/2014	Chen	359/462	G02B 30/31
2015/0334369	12/2014	Bruls	348/43	H04N 13/128
2016/0070112	12/2015	Song	359/471	G02B 30/40
2016/0077349	12/2015	An	359/462	H04N 13/351
2016/0150209	12/2015	Dadash Pour	348/51	H04N 13/128
2016/0191904	12/2015	An	348/51	G02B 30/27
2016/0219258	12/2015	Woodgate	N/A	H04N 23/661
2016/0234482	12/2015	Bickerstaff	N/A	H04N 13/128
2016/0241842	12/2015	Newbery	N/A	G03B 35/10
2016/0360187	12/2015	Smithwick	N/A	G02B 30/27
2017/0038598	12/2016	Song	N/A	G02B 30/36
2017/0041596	12/2016	Park	N/A	H04N 13/305
2017/0078652	12/2016	Hua	N/A	G06V 20/20
2017/0289518	12/2016	Kim	N/A	H04N 13/239
2017/0366797	12/2016	Kim	N/A	H04N 13/128
2018/0007349	12/2017	Lee	N/A	G02B 30/27
2019/0180514	12/2018	Sasaki	N/A	H04N 13/128
2020/0029067	12/2019	Lee	N/A	H04N 13/383
2020/0126494	12/2019	Chen	N/A	G09G 5/10
2020/0201036	12/2019	Hong	N/A	B60R 1/31
2021/0132693	12/2020	Pulli	N/A	G02B 27/0093
2021/0136354	12/2020	Valli	N/A	G02B 27/017
2021/0314542	12/2020	Jung	N/A	H04N 13/122
2022/0087813	12/2021	Burge	N/A	G02C 7/104
2022/0264076	12/2021	Makinen	N/A	G02B 30/26
2022/0277522	12/2021	Katsuki	N/A	H04N 17/00
2022/0385880	12/2021	Nims	N/A	H04N 13/271
2024/0118550	12/2023	Lai	N/A	G02B 27/0172
2024/0169673	12/2023	Xiong	N/A	G06T 17/205
2024/0187561	12/2023	Tomizawa	N/A	H04N 13/32
2024/0221276	12/2023	Foda	N/A	H04N 13/128
2024/0414309	12/2023	He	N/A	H04N 13/279

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104781873	12/2014	CN	N/A
113315962	12/2020	CN	N/A
114387324	12/2021	CN	N/A
201518847	12/2014	TW	N/A
201815162	12/2017	TW	N/A
202201348	12/2021	TW	N/A

Primary Examiner: Beard; Charles L

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 111150471, filed on Dec. 28, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a method and device for adjusting an image, and particularly relates to a method and device for adjusting a depth of a stereoscopic image.

Description of Related Art

(3) The interpupillary distance (IPD) of each person is quite different. For people with a large IPD, a stereoscopic feeling when seeing ordinary objects is more prominent compared with people with a small IPD. For this reason, when using a stereoscopic display product, such as a virtual reality display (VR display), an augmented reality display (AR display), a stereoscopic TV, inputting the IPD of a seer or a user into a software is required to achieve a realistic stereoscopic size and feeling. As in the stereoscopic TV field, image sources photographed by different users have different depth records. When another user sees through a stereoscopic display, since two users have different IPDs, perceived depths are also different.

SUMMARY

(4) The disclosure provides a method for adjusting a depth of a stereoscopic image, which enables users with different interpupillary distances (IPDs), or stereoscopic image photographers and stereoscopic display viewers all to see stereoscopic images with the same depth effect.

(5) The disclosure provides a device for adjusting a depth of a stereoscopic image, which enables users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers all to see stereoscopic images with the same depth effect.

(6) An embodiment of the disclosure provides a method for adjusting a depth of a stereoscopic image, which includes the following. A first disparity correction amount is obtained according to a distance between centers of two lenses of a stereoscopic camera and an IPD of a first observer, in which a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object. A second disparity correction amount is obtained according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, in which a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera. A third disparity correction amount is obtained according to the IPD of the first observer and an IPD of a second observer, in which a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object. A stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is transmitted to the stereoscopic display.

(7) An embodiment of the disclosure provides a device for adjusting a depth of a stereoscopic image, which includes at least one controller used to receive a stereoscopic image signal from a stereoscopic camera and to control a display content of a stereoscopic display. The controller is configured to execute the following: obtain a first disparity correction amount according to a distance between centers of two lenses of the stereoscopic camera and an IPD of a first observer, in which a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of

a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object; obtain a second disparity correction amount according to a difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, in which a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera; obtain a third disparity correction amount according to the IPD of the first observer and an IPD of a second observer, in which a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and transmit a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the stereoscopic display.

(8) An embodiment of the disclosure provides a method for adjusting a depth of a stereoscopic image, which includes the following. A distance between centers of two lenses of a stereoscopic camera is adjusted to be the same as an IPD of a first observer, so that a stereoscopic image distance of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object. A first disparity correction amount is obtained according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, in which a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera. A second disparity correction amount is obtained according to the IPD of the first observer and an IPD of a second observer, in which a stereoscopic image distance formed after adding the first disparity correction amount and the second disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object. A stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount and the second disparity correction amount is transmitted to the stereoscopic display.

(9) In the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the first disparity correction amount is obtained according to the distance between centers of the two lenses of the stereoscopic camera and the IPD of the first observer, the second disparity correction amount is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, the third disparity correction amount is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect. In the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the distance between centers of the two lenses of the stereoscopic camera is adjusted to be the same as the IPD of the first observer, the first disparity correction amount is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, the second disparity correction amount is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount and the second disparity correction amount is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
- (2) FIG. 1A, FIG. 1B, and FIG. 1C are schematic diagrams of a disparity on a stereoscopic display.
- (3) FIG. 2 is a flowchart of a method for adjusting a depth of a stereoscopic image according to an embodiment of the disclosure.
- (4) FIG. 3A is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image in FIG. 2.
- (5) FIG. 3B is a schematic diagram of another side of the stereoscopic camera in FIG. 3A.
- (6) FIG. 4 and FIG. 5 are graphs used to calculate a disparity correction amount $\delta_{\text{sub.A}}$ in FIG. 2.
- (7) FIG. 6, FIG. 7, and FIG. 8 are graphs used to calculate a disparity correction amount $\delta_{\text{sub.B}}$ in FIG. 2.
- (8) FIG. 9 is a graph for calculating a disparity correction amount $\delta_{\text{sub.C}}$ in FIG. 2.
- (9) FIG. 10 is a flowchart of a method for adjusting a depth of a stereoscopic image according to another embodiment of the disclosure.
- (10) FIG. 11 is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image shown in FIG. 10.

DESCRIPTION OF THE EMBODIMENTS

- (11) FIG. 1A, FIG. 1B, and FIG. 1C are schematic diagrams of a disparity on a stereoscopic display. Please refer to FIG. 1A, FIG. 1B and FIG. 1C. A depth of a stereoscopic image is mainly displayed by using a difference amount of a disparity **51** of two eyes, and the depth mentioned here is a distance between the stereoscopic image and the eyes felt by an observer. An image **52** and an image **54** are images projected to the left eye and the right eye, respectively. When the disparity **51** is large, a stereoscopic feeling generated is strong, which means that the stereoscopic image stands out (that is, the depth is small); on the contrary, when the disparity **51** is small, a stereo vision perception is minor, which means that the depth is large and the stereoscopic image is far away from a user.
- (12) If adjusting the depth of the stereoscopic image is desired, the difference amount of a disparity of two eyes may be used to increase a disparity between the image **52** and the image **54** (such as adjusting from the disparity **51** in FIG. 1A to a larger disparity **51'** in FIG. 1B) so that the stereoscopic image may be made to stand out and be close to the user. On the other hand, as in FIG. 1C, if a disparity **51''** is adjusted to 0, the depth of the stereoscopic image is on a display surface of the stereoscopic display.
- (13) Therefore, by adjusting the amount of the disparity, the depth of the stereoscopic image may be adjusted.
- (14) FIG. 2 is a flowchart of a method for adjusting a depth of a stereoscopic image according to an embodiment of the disclosure. FIG. 3A is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image in FIG. 2. FIG. 3B is a schematic diagram of another side of a stereoscopic camera in FIG. 3A. Please refer to FIG. 2 and FIG. 3A first. This embodiment provides a device for adjusting a depth of a stereoscopic image **100**, which includes at least one controller **110** (one of the at least one controller **110** is taken as an example in FIG. 3A) used to receive a stereoscopic image signal from a stereoscopic camera **200** and to control a display content of a stereoscopic display **120**. The controller **110** is configured to execute Step **S110** to Step **S140** in FIG. 2. In other words, the method for adjusting a depth of a stereoscopic image of this embodiment includes Step **S110** to Step **S140**, and it may be implemented by the device for adjusting a depth of the stereoscopic image **100** of this embodiment.
- (15) In the method for adjusting a depth of a stereoscopic image of this embodiment, Step **S110** is

first executed as follows. A disparity correction amount $\delta_{\text{sub.A}}$ is obtained according to a distance B between centers of two lenses **210** of the stereoscopic camera **200** and an interpupillary distance (IPD) IPD.sub.1 of a first observer **60**, in which a stereoscopic image distance (that is, a distance between a stereoscopic image and eyes perceived by a user when seeing the stereoscopic image, that is, a depth of the stereoscopic image) formed after adding the disparity correction amount $\delta_{\text{sub.A}}$ to a disparity of a stereoscopic image of an object **80** photographed by the stereoscopic camera **200** is equal to a distance D1 (that is, an actual distance between eyes **62** of the first observer **60** and the object **80**) of the first observer **60** seeing the object **80**.

(16) Next, Step S120 is executed as follows. A disparity correction amount $\delta_{\text{sub.B}}$ is obtained according to a difference between image distances of the stereoscopic display **120** and the stereoscopic camera **200** at zero disparity, in which a stereoscopic image distance formed after adding the disparity correction amount $\delta_{\text{sub.B}}$ to a disparity of a stereoscopic image displayed by the stereoscopic display **120** is equal to a stereoscopic image distance provided by the stereoscopic camera **200**.

(17) Then, Step S130 is executed as follows. A disparity correction amount $\delta_{\text{sub.C}}$ is obtained according to the IPD IPD.sub.1 of the first observer **60** and an IPD IPD.sub.2 (that is, a distance between pupils of two eyes **72** of a second observer **70**) of the second observer **70**, in which a stereoscopic image distance formed after adding the disparity correction amount $\delta_{\text{sub.A}}$, the disparity correction amount $\delta_{\text{sub.B}}$, and the disparity correction amount $\delta_{\text{sub.C}}$ to the disparity of the stereoscopic image displayed by the stereoscopic display **120** is equal to the distance D1 of the first observer **60** seeing the object **80**.

(18) Afterward, Step S140 is executed as follows. A stereoscopic image of the object **80** photographed by the stereoscopic camera **200** which has been corrected by the disparity correction amount $\delta_{\text{sub.A}}$, the disparity correction amount $\delta_{\text{sub.B}}$, and the disparity correction amount $\delta_{\text{sub.C}}$ is transmitted to the stereoscopic display **120**.

(19) In this embodiment, the stereoscopic camera **200** may be integrated with a virtual reality display **300** to display the stereoscopic image of the object **80** photographed by the stereoscopic camera **200**. For example, the front of the stereoscopic camera **200** is provided with the lens **210**, which is able to photograph the object **80**, and the back of the stereoscopic camera **200** is provided with virtual image displays **220**, and the observer **60** is able to see a stereoscopic image generated by virtual reality by aligning the two eyes **62** to the two virtual image displays **220** respectively. The method for adjusting a depth of a stereoscopic image further includes the following. A stereoscopic image of the object **80** photographed by the stereoscopic camera **200** which has been corrected by the disparity correction amount $\delta_{\text{sub.A}}$ of the stereoscopic image is displayed by the virtual reality display **300**, so that a stereoscopic image distance of a stereoscopic image seen by the first observer **60** through the virtual reality display **300** is equal to the distance D1 of the first observer **60** directly seeing the object **80**.

(20) In this embodiment, the disparity correction amount $\delta_{\text{sub.A}}$, the disparity correction amount $\delta_{\text{sub.B}}$, and the disparity correction amount $\delta_{\text{sub.C}}$ may be calculated by a pixel difference, and the pixel difference refers to a distance of the number of pixels deviated. In addition, in this embodiment, the method of obtaining the disparity correction amount $\delta_{\text{sub.A}}$, the disparity correction amount $\delta_{\text{sub.B}}$, and the disparity correction amount $\delta_{\text{sub.C}}$ is performed through calculation with triangulation. The calculation process of Steps S110, S120, and S130 will be described in detail below.

(21) FIG. 4 and FIG. 5 are graphs used to calculate the disparity correction amount $\delta_{\text{sub.A}}$ of FIG. 2. Please refer to FIG. 3A, FIG. 4, and FIG. 5. When calculating the disparity correction amount $\delta_{\text{sub.A}}$, assuming that the first observer **60** utilizes the stereoscopic camera **200** to take a picture or record a video to generate a stereoscopic image file, but due to the IPD IPD.sub.1 of the first observer **60** and the distance B between centers of the two lenses **210** of the stereoscopic camera **200** is different, so the disparity correction amount $\delta_{\text{sub.A}}$ needs to be added at this time. Also, the following mathematical formula is used to calculate.

$$(22) \quad Z_c = \frac{fB}{(x_c - x'_c)} = \frac{fB}{\frac{1}{m_c}(u_L - u_R)} \quad (\text{Formula1}) \quad m_c = \frac{\text{camera_pixels}}{\text{mm}} \quad (\text{Formula2})$$

(23) FIG. 4 is a simple structure diagram of the stereoscopic camera **200**, B is the distance between centers of the two lenses **210** of the stereoscopic camera **200**, X is a certain object point photographed, x and x' are actual projection positions on a photosensitive element of a camera, f is a focal length of the camera, and m.sub.c is a pixel density. For example, camera_pixels in Formula 2 is the number of pixels per millimeter (mm) on the photosensitive element. As in Formula 1, a depth Z.sub.c is composed of a pixel position difference (u.sub.L-u.sub.R) projected by X on a photo. That is, the depth may be adjusted by adjusting the pixel difference between the left eye and the right eye.

$$(24) \quad Z_c = \frac{B}{\tan \frac{c}{2}} \quad (\text{Formula3}) \quad \tan \frac{c}{2} = \frac{(x_c - x'_c)}{2f} = \frac{\frac{1}{m_c}(u_L - u_R)}{2f} \quad (\text{Formula4}) \quad Z_1 = \frac{IPD_1}{\tan \frac{1}{2}} \quad (\text{Formula5})$$

$$Z_c = Z_1 \quad (\text{Formula6}) \quad 1 = c + \quad (\text{Formula7}) \quad \tan \frac{1}{2} = \frac{\frac{IPD_1}{B} - 1}{\frac{1}{\tan \frac{c}{2}} + \frac{IPD_1}{B} \tan \frac{c}{2}} \quad (\text{Formula8})$$

$$\text{Disparity}_1 = (u_L - u_R + \quad (\text{Formula9}) \quad A = PPD_{\text{cm}} \cdot \text{Math.} \quad 1 \quad (\text{Formula10})$$

(25) Assuming that the stereoscopic camera **200** is integrated with the virtual reality display **300**, the first observer **60** sees a content photographed by oneself through the virtual reality display **300**. The IPD IPD.sub.1 of the first observer **60** is different from B, as in FIG. 5. Since a stereoscopic image distance is generated by a disparity, and a virtual image imaging distance of the virtual reality display **300** may be infinite, it is necessary to convert the unit of the disparity from mm into an angle deg. A difference amount of the disparity is φ1, as in Formula 7. Also, the purpose is that a depth Z.sub.1 seen by the first observer **60** is equal to the depth Z.sub.c photographed. From Formula 3 and Formula 5, Formula 8 may be obtained to obtain φ1. In addition, φ1 is converted into a pixel difference (i.e., the disparity correction amount δ.sub.A), as in Formula 10, and the conversion of the angle into the pixel is calculated by the pixel per deg PPD.sub.cm of the virtual reality display **300**. In this way, a difference amount of the pixel difference δ.sub.A may be increased. That is, a depth photographed by the stereoscopic camera **200** is the same as a depth seen by the first observer **60**. A new disparity Disparity.sub.1 (i.e., (u.sub.1L-u.sub.1R)) is described as in Formula 9.

(26) FIG. 6, FIG. 7, and FIG. 8 are graphs used to calculate the disparity correction amount δ.sub.B in FIG. 2. Please refer to FIG. 3A, FIG. 6, FIG. 7, and FIG. 8. When calculating the disparity correction amount δ.sub.A, as illustrated in FIG. 6, since placements of the virtual reality display **300** and the stereoscopic camera **200** are parallel placements, under such a structure, when the disparity is 0, the equivalent is a two-dimensional plane **302**, and the two-dimensional plane **302** is at infinity. However, since when the disparity of the stereoscopic display **120** is 0, when the equivalent is a two-dimensional plane, the two-dimensional plane is not at infinity, but on a display surface **122** of the display, and a distance is d, so in a situation of the same disparity amount, when images at other positions or at different depths are projected on the stereoscopic display **120**, the distance also changes, like as changing from infinity to d when the disparity is zero. Therefore, the disparity correction amount δ.sub.B needs to be added. That is to say, the image distance of the stereoscopic camera **200** at zero disparity is infinite, and the image distance of the stereoscopic display **120** at zero disparity is a finite value. Step S120 includes calculating the disparity correction amount δ.sub.B according to the finite value.

$$(27) \quad m_m = \frac{\text{monitor_pixels}}{\text{mm}} \quad (\text{Formula11}) \quad \frac{Z_m}{IPD_1} = \frac{d - Z_m}{\frac{1}{m_m}(u_{1L} - u_{1R} + B)} \quad (\text{Formula12})$$

$$Z_m = \frac{d}{\frac{1}{m_m}(u_{1L} - u_{1R} + B) + 1} \quad (\text{Formula13}) \quad Z_m = Z_1 \quad (\text{Formula14})$$

$$\frac{d}{\frac{1}{m_m}(u_{1L} - u_{1R} + B) + 1} = \frac{IPD_1}{\tan \frac{(u_L - u_R)}{2PPD_{\text{cm}}}} \quad (\text{Formula15})$$

$$B = m_m d \tan \frac{(u_{1L} - u_{1R})}{2PPD_{\text{cm}}} - m_m IPD_1 - (u_{1L} - u_{1R}) \quad (\text{Formula16})$$

$$\text{Disparity}_m = (u_{1L} - u_{1R} + \quad (\text{Formula17})$$

(28) As in FIG. 8, δ.sub.B is added to a disparity content projected on the stereoscopic display **120**. In

addition, the pixel density of the stereoscopic display **120** is $m_{\text{sub.m}}$, and monitor_pixels is the number of pixels per mm of the stereoscopic display **120**. Assuming that the observer is the same, that is, the first observer **60** sees the stereoscopic display **120**, and a resulting stereoscopic depth is $Z_{\text{sub.m}}$. From Formula 14, of which the objective is that a seen depth is equal to the depth when seeing the virtual reality display **300**, and then the difference amount $\delta_{\text{sub.B}}$ is obtained. (29) Formula 12 may be obtained from similar triangles, and $Z_{\text{sub.m}}$ may be obtained after arrangement, as Formula 13. Formula 15 may be obtained from Formula 14, and $\delta_{\text{sub.B}}$ may be obtained after arrangement, as Formula 16. A new disparity amount Disparity.sub.m (i.e., $(u_{\text{sub.mL}} - u_{\text{sub.mR}})$) is as described in Formula 17.

(30) FIG. 9 is a graph for calculating the disparity correction amount $\delta_{\text{sub.C}}$ in FIG. 2. Please refer to FIG. 3A and FIG. 9. When calculating the disparity correction amount $\delta_{\text{sub.C}}$, assuming that it is the second observer **70** who sees the stereoscopic display **120**, and since the second observer **70** and the first observer **60** have different IPDs, which are $IPD_{\text{sub.2}}$ and $IPD_{\text{sub.1}}$ respectively, the correction amount $\delta_{\text{sub.C}}$ needs to be added.

$$(31) \quad \frac{Z_2}{IPD_2} = \frac{d - Z_2}{\frac{1}{m_m}(u_{mL} - u_{mR} + c)} \quad (\text{Formula18}) \quad Z_m = \frac{d}{\frac{1}{m_m}(u_{mL} - u_{mR} + c) + 1} \quad (\text{Formula19}) \quad Z_2 = Z_m \quad (\text{Formula20})$$

$$\frac{d}{\frac{1}{m_m}(u_{mL} - u_{mR} + c) + 1} = \frac{d}{\frac{1}{m_m}(u_{mL} - u_{mR}) + 1} \quad (\text{Formula21}) \quad c = \left(\frac{IPD_2}{IPD_1} - 1\right)(u_{mL} - u_{mR}) \quad (\text{Formula22})$$

$$\text{Disparity}_2 = (u_{mL} - u_{mR} + c) \quad (\text{Formula23})$$

(32) As in FIG. 12, a $\delta_{\text{sub.C}}$ is added and the IPD of the second observer **70** is $IPD_{\text{sub.2}}$, and a stereoscopic visual distance is $Z_{\text{sub.2}}$. Formula 18 may be obtained from similar triangles, and $Z_{\text{sub.2}}$ may be obtained after arrangement, as Formula 19. Since the object is for the second observer **70** and the first observer **60** to feel the same stereoscopic depth, as in Formula 20, Formula 21 may be obtained, and Formula 22 may be obtained after arrangement. The increased amount of the disparity is $\delta_{\text{sub.C}}$. A new disparity Disparity.sub.2 (i.e. $(u_{\text{sub.2L}} - u_{\text{sub.2R}})$) is as described in Formula 23.

(33) After a series of derivations, $\delta_{\text{sub.A}}$, $\delta_{\text{sub.B}}$, and $\delta_{\text{sub.C}}$ may be obtained. Then, the disparity of the second observer **70** may be obtained by adding δ_A , $\delta_{\text{sub.B}}$, and $\delta_{\text{sub.C}}$ to the disparity photographed by the first observer **60**, and the stereoscopic depth seen by the two observers may be the same.

(34) In an embodiment, the controller **110** is, for example, a central processing unit (CPU), a microprocessor, a digital signal processor (DSP), a programmable controller, a programmable logic device (PLD) or other similar devices or a combination of the devices, and the disclosure is not limited thereto. In addition, in an embodiment, each function of the controller **110** may be implemented as a plurality of program codes. These program codes are stored in a memory and are executed by the controller **110**. Alternatively, in an embodiment, each function of the controller **110** may be implemented as one or more circuits. The disclosure does not limit the implementation of the functions of the controller **110** by means of software or hardware.

(35) In this embodiment, the device for adjusting a depth of a stereoscopic image **100** may be a notebook computer, and the controller **110** is, for example, the CPU of the notebook computer, and the stereoscopic display **120** is, for example, the display of the notebook computer, but the disclosure is not limited thereto. In other embodiments, the controller **110** may also be separated from the stereoscopic display **120** and be positioned on another server or desktop computer. In addition, there may be a plurality of controllers **110**, and a controller performing Step S110 may be positioned on the stereoscopic camera **200**, while a controller performing Steps S120 to S140 may be positioned on a notebook computer, a server or a desktop computer.

(36) FIG. 10 is a flowchart of a method for adjusting a depth of a stereoscopic image according to another embodiment of the disclosure, and FIG. 11 is a schematic diagram illustrating a part of a flow of the method for adjusting a depth of a stereoscopic image in FIG. 10. Please refer to FIG. 10 and FIG. 11. The method for adjusting a depth of a stereoscopic image of this embodiment is similar to the method for adjusting a depth of a stereoscopic image in FIG. 2, while a difference is as the

following. In this embodiment, a distance B' between centers of the two lenses **210** of the stereoscopic camera **200** is adjustable, for example, may be adjusted by a mechanism. In Step **S110'** of the method for adjusting a depth of a stereoscopic image of this embodiment, the distance B' between centers of the two lenses **210** of the stereoscopic camera **200** is adjusted to be the same as the IPD $IPD.sub.1$ of the first observer **60**, so that a stereoscopic image distance of an object photographed by the stereoscopic camera **200** is equal to the distance $D1$ of the first observer **60** seeing the object. In this way, after such adjustment, it may be considered that the disparity correction amount $\delta.sub.A$ is zero. In this case, in Step **S140'**, there is no need to consider the disparity correction amount $\delta.sub.A$, so step **S140'** may be to transmit a stereoscopic image of the object **80** photographed by the stereoscopic camera **200** which has been corrected by the disparity correction amount $\delta.sub.B$ and the disparity correction amount $\delta.sub.C$ to the stereoscopic display **120**. In addition, the device for adjusting a depth of a stereoscopic image **100** of the embodiment of the disclosure may also be used to implement the method for adjusting a depth of a stereoscopic image shown in FIG. **10**.

(37) In summary, in the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the disparity correction amount $\delta.sub.A$ is obtained according to the distance between centers of the two lenses of the stereoscopic camera and the IPD of the first observer, the disparity correction amount $\delta.sub.B$ is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, the disparity correction amount $\delta.sub.C$ is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the disparity correction amount $\delta.sub.A$, the disparity correction amount $\delta.sub.B$, and the disparity correction amount $\delta.sub.C$ is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect. In the method for adjusting a depth of a stereoscopic image and the device for adjusting a depth of a stereoscopic image of the embodiments of the disclosure, the distance between centers of the two lenses of the stereoscopic camera is adjusted to be the same as the IPD of the first observer, the disparity correction amount $\delta.sub.B$ is obtained according to the difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, a disparity correction amount $\delta.sub.C$ is obtained according to the IPD of the first observer and the IPD of the second observer, and the stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the disparity correction amount $\delta.sub.B$ and the disparity correction amount $\delta.sub.C$ is transmitted to the stereoscopic display. Therefore, users with different IPDs, or stereoscopic image photographers and stereoscopic display viewers are all enabled to see stereoscopic images with the same depth effect.

Claims

1. A method for adjusting a depth of a stereoscopic image, comprising: obtaining a first disparity correction amount according to a distance between centers of two lenses of a stereoscopic camera and an interpupillary distance (IPD) of a first observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object; obtaining a second disparity correction amount according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, wherein a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera; obtaining a third disparity correction amount according to the IPD of the first observer and an IPD of a second observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity

correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and transmitting a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the stereoscopic display.

2. The method for adjusting the depth of the stereoscopic image according to claim 1, further comprising calculating the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount by a pixel difference.

3. The method for adjusting the depth of the stereoscopic image according to claim 1, wherein the stereoscopic camera is integrated with a virtual reality display to display the stereoscopic image of the object photographed by the stereoscopic camera, wherein the method for adjusting the depth of the stereoscopic image further comprises displaying, by the virtual reality display, a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, so that a stereoscopic image distance of the stereoscopic image seen by the first observer through the virtual reality display is equal to a distance of the first observer directly seeing the object.

4. The method for adjusting the depth of the stereoscopic image according to claim 1, wherein the image distance of the stereoscopic camera at zero disparity is infinite, and the image distance of the stereoscopic display at zero disparity is a finite value.

5. The method for adjusting the depth of the stereoscopic image according to claim 4, wherein obtaining the second disparity correction amount according to the difference between the image distances of the stereoscopic display and the stereoscopic camera at zero disparity comprises calculating the second disparity correction amount according to the finite value.

6. The method for adjusting the depth of the stereoscopic image according to claim 1, wherein the method for obtaining the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is performed through calculation with triangulation.

7. A device for adjusting a depth of a stereoscopic image, comprising: at least one controller configured to receive a stereoscopic image signal from a stereoscopic camera and to control a display content of a stereoscopic display, wherein the controller is configured to: obtain a first disparity correction amount according to a distance between centers of two lenses of the stereoscopic camera and an IPD of a first observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object; obtain a second disparity correction amount according to a difference between image distances of the stereoscopic display and the stereoscopic camera at zero disparity, wherein a stereoscopic image distance formed after adding the second disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera; obtain a third disparity correction amount according to the IPD of the first observer and an IPD of a second observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and transmit a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount to the stereoscopic display.

8. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the controller is further configured to: calculate the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount by a pixel difference.

9. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the stereoscopic camera is integrated with a virtual reality display to display the stereoscopic image of the object photographed by the stereoscopic camera, wherein the controller is further configured to:

display, by the virtual reality display, a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount, so that a stereoscopic image distance of the stereoscopic image seen by the first observer through the virtual reality display is equal to a distance of the first observer directly seeing the object.

10. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the image distance of the stereoscopic camera at zero disparity is infinite, and the image distance of the stereoscopic display at zero disparity is a finite value.

11. The device for adjusting the depth of the stereoscopic image according to claim 10, wherein obtaining the second disparity correction amount according to the difference between the image distances of the stereoscopic display and the stereoscopic camera at zero disparity comprises calculating the second disparity correction amount according to the finite value.

12. The device for adjusting the depth of the stereoscopic image according to claim 7, wherein the method for obtaining the first disparity correction amount, the second disparity correction amount, and the third disparity correction amount is performed through calculation with triangulation.

13. A method for adjusting a depth of a stereoscopic image, comprising: adjusting a distance between centers of two lenses of a stereoscopic camera to be the same as an IPD of a first observer, so that a stereoscopic image distance of an object photographed by the stereoscopic camera is equal to a distance of the first observer seeing the object; obtaining a first disparity correction amount according to a difference between image distances of a stereoscopic display and the stereoscopic camera at zero disparity, wherein a stereoscopic image distance formed after adding the first disparity correction amount to a disparity of a stereoscopic image displayed by the stereoscopic display is equal to a stereoscopic image distance provided by the stereoscopic camera; obtaining a second disparity correction amount according to the IPD of the first observer and an IPD of a second observer, wherein a stereoscopic image distance formed after adding the first disparity correction amount and the second disparity correction amount to the disparity of the stereoscopic image displayed by the stereoscopic display is equal to the distance of the first observer seeing the object; and transmitting a stereoscopic image of the object photographed by the stereoscopic camera which has been corrected by the first disparity correction amount and the second disparity correction amount to the stereoscopic display.

14. The method for adjusting the depth of the stereoscopic image according to claim 13, further comprising calculating the first disparity correction amount and the second disparity correction amount by a pixel difference.

15. The method for adjusting the depth of the stereoscopic image according to claim 13, wherein the stereoscopic camera is integrated with a virtual reality display to display the stereoscopic image of the object photographed by the stereoscopic camera.

16. The method for adjusting the depth of the stereoscopic image according to claim 13, wherein the image distance of the stereoscopic camera at zero disparity is infinite, and the image distance of the stereoscopic display at zero disparity is a finite value.

17. The method for adjusting the depth of the stereoscopic image according to claim 16, wherein obtaining the first disparity correction amount according to the difference between the image distances of the stereoscopic display and the stereoscopic camera at zero disparity comprises calculating the first disparity correction amount according to the finite value.

18. The method for adjusting the depth of the stereoscopic image according to claim 13, wherein the method for obtaining the first disparity correction amount and the second disparity correction amount is performed through calculation with triangulation.
